



PART C DMX Patch & Step-by-Step Guide

The venue patch (40 fixtures + optional pinspots), channel maps, Art-Net/sACN network, fader map, the patch procedure for the show file, rig sheets, power and troubleshooting.

Revision R13.1 · 28 September 2026

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DMX Patch & Step-by-Step Guide

Confirmed desk patch, channel maps, network, and the procedure from load to rig check

40 fixtures patched (base file)	2 optional pinspots #41–42	2 universes (DMX + Art-Net/sACN)	0 re-addressing needed
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WHERE THINGS LIVE
TLM_SHOW_2026_R13_FLASHY_SCENE_SPLIT.mtr carries the venue base patch, custom fixtures and network. **HAZE is P5 M4**, P2 M3 is the ship/storm scene memory, and the optional pinspots go on **P5 M7** (free).

THE ADDRESS TRAVELS WITH THE FIXTURE
Every fixture keeps its desk number and DMX address wherever it hangs. Moving a C42 to the FOH bar means re-cabling it into the universe 2 line — not changing its address.

Address map

Universe	Range	Fixtures	Free
U1	1–36	PixBars #33–38 (6 ch)	
U1	37–38	Pinspots #41–42 (optional)	
U1	39–300	—	262 free (future moving heads)
U1	301–420	Tour Pro Zooms #13–22 (12 ch)	
U1	421–486	TourCOB PARs #23–32 and #39 (6 ch)	
U1	487–488	Hazer #40 (2 ch)	489–512 free
U2	101–232	Lightsky C42 #1–12 (11 ch)	1–100 and 233–512 free

1 Master DMX patch

#	Desk library	Mode	Universe:address	Intensity at	Display	Position	Fader
1	LIGHTSKY · CX 42 NEW	11 ch	U2:101–111	109	A101	FOH 1	Console
2	LIGHTSKY · CX 42 NEW	11 ch	U2:112–122	120	A112	FOH 2	Console
3	LIGHTSKY · CX 42 NEW	11 ch	U2:123–133	131	A123	FOH 3	Console
4	LIGHTSKY · CX 42 NEW	11 ch	U2:134–144	142	A134	FOH 4	Console
5	LIGHTSKY · CX 42 NEW	11 ch	U2:145–155	153	A145	FOH 5	Console
6	LIGHTSKY · CX 42 NEW	11 ch	U2:156–166	164	A156	FOH 6	Console
7	LIGHTSKY · CX 42 NEW	11 ch	U2:167–177	175	A167	FOH 7	Console
8	LIGHTSKY · CX 42 NEW	11 ch	U2:178–188	186	A178	FOH 8	Console
9	LIGHTSKY · CX 42 NEW	11 ch	U2:189–199	197	A189	FOH 9	Console
10	LIGHTSKY · CX 42 NEW	11 ch	U2:200–210	208	A200	FOH 10	Console
11	LIGHTSKY · CX 42 NEW	11 ch	U2:211–221	219	A211	FOH 11	Console
12	LIGHTSKY · CX 42 NEW	11 ch	U2:222–232	230	A222	FOH 12	Console
13	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:301–312	301	A301	LX1	Console
14	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:313–324	313	A313	LX1	Console
15	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:325–336	325	A325	LX1	Console
16	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:337–348	337	A337	LX1	Console
17	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:349–360	349	A349	LX1	Console
18	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:361–372	361	A361	LX1	Console
19	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:373–384	373	A373	LX2	Console
20	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:385–396	385	A385	LX2	Console
21	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:397–408	397	A397	SR boom	Console
22	TOUR PR0 · ZOOM 12 CHANNEL	12 ch	U1:409–420	409	A409	SL boom	Console
23	TOUR PRO · TOURCOB PAR	6 ch	U1:421–426	421	A421	LX2	Console
24	TOUR PRO · TOURCOB PAR	6 ch	U1:427–432	427	A427	LX2	Console
25	TOUR PRO · TOURCOB PAR	6 ch	U1:433–438	433	A433	LX2	Wing 1

#	Desk library	Mode	Universe:address	Intensity at	Display	Position	Fader
26	TOUR PRO · TOURCOB PAR	6 ch	U1:439–444	439	A439	LX2	Wing 1
27	TOUR PRO · TOURCOB PAR	6 ch	U1:445–450	445	A445	LX2	Wing 1
28	TOUR PRO · TOURCOB PAR	6 ch	U1:451–456	451	A451	LX2	Wing 1
29	TOUR PRO · TOURCOB PAR	6 ch	U1:457–462	457	A457	SR boom	Wing 1
30	TOUR PRO · TOURCOB PAR	6 ch	U1:463–468	463	A463	SR boom	Wing 1
31	TOUR PRO · TOURCOB PAR	6 ch	U1:469–474	469	A469	SL boom	Wing 1
32	TOUR PRO · TOURCOB PAR	6 ch	U1:475–480	475	A475	SL boom	Wing 1
33	TOUR PRO · PIXBAR 6CH	6 ch	U1:1–6	virtual	A001	LX2	Wing 1
34	TOUR PRO · PIXBAR 6CH	6 ch	U1:7–12	virtual	A007	LX2	Wing 1
35	TOUR PRO · PIXBAR 6CH	6 ch	U1:13–18	virtual	A013	LX2	Wing 1
36	TOUR PRO · PIXBAR 6CH	6 ch	U1:19–24	virtual	A019	LX2	Wing 1
37	TOUR PRO · PIXBAR 6CH	6 ch	U1:25–30	virtual	A025	LX2	Wing 2
38	TOUR PRO · PIXBAR 6CH	6 ch	U1:31–36	virtual	A031	LX2	Wing 2
39	TOUR PRO · TOURCOB PAR	6 ch	U1:481–486	481	A481	LX2	Wing 2
40	GENERIC · HAZER 2CH	2 ch	U1:487–488	487	A487	Floor, US	Wing 2
41	Generic dimmer (1 ch)	1 ch	U1:37	37	A037	FOH SR end	Wing 2
42	Generic dimmer (1 ch)	1 ch	U1:38	38	A038	FOH SL end	Wing 2

2 Channel maps

Each fixture must be in the matching DMX mode (11 / 12 / 6 channels) or colours land on the wrong channels. Home = value when the fixture is cleared.

Lightsky 8800-C42 · CX 42 NEW · 11 ch

Ch	Function	Home / rule
1	Red	255
2	Amber	0
3	Lime	0
4	Green	255
5	Blue	255
6	UV	0
7	CCT	0
8	Colour macro	0 · keep 0
9	Intensity	0
10	Control	0 · keep 0
11	Strobe	0 · keep 0

Tour Pro Zoom · ZOOM 12 CHANNEL · 12 ch

Ch	Function	Home / rule
1	Intensity	0
2	Red	255
3	Green	255
4	Blue	255
5	White	0
6	Control	keep 0
7	Strobe	keep 0
8	Zoom	set in focus
9	Other	keep 0
10	Default	keep 0
11	Auto speed	keep 0
12	Reset	keep 0

Tour Pro TourCOB PAR · 6 ch

Ch	Function	Home / rule
1	Intensity	0
2	Red	255
3	Green	255
4	Blue	255
5	White	0
6	Strobe	keep 0

Tour Pro PixBar · PIXBAR 6CH · 6 ch

Ch	Function	Home / rule
1	Red	255
2	Green	255
3	Blue	255
4	White	0
5	Amber	0
6	UV	0

Hazer · HAZER 2CH · 2 ch

Ch	Function	Home / rule
1	Output	0
2	Fan	0

FOH C42s RUN FROM DIMMERS — SET NON-DIM

All 12 C42s stay on the FOH bar (approved: the bar runs back to the dimmer setup). Set every dimmer channel feeding them to non-dim / hard power at the rack before power-up, spread 4 C42 per 10 A circuit (1,200 W each). If the rack also listens to DMX, its channels must never be faded: at 0 the C42s lose power. Their brightness is set by their own intensity channel (ch 9), not by the dimmer.

KEEP CONTROL-TYPE CHANNELS AT 0

Control, Strobe, Reset, Auto speed, Other, Default and Colour macro can trigger resets, strobing or built-in programs. Every cue in the R13.1 file keeps them at 0; lightning is done with intensity (R-04). The Zoom library name is spelled TOUR PRO (zero) on the desk — cosmetic only.

3 DMX runs, network and fader map

Line	Route	
Universe 1 (desk XLR)	Desk → LX1 Zooms #13–18 → LX2 (#19–20, 23–28, 39, 33–38) → SR boom (#21, 29–30) → SL boom (#22, 31–32) → hazer #40 · terminate	
Universe 2 (Ethernet)	Desk → switch → node 2.0.0.10 → FOH bar C42 #1–12 · terminate	
Setting	Desk (Mantra)	Node for universe 2
IP address	2.0.0.1 (static, DHCP off)	2.0.0.10
Subnet mask	255.0.0.0	255.0.0.0
Protocols	Art-Net on · sACN on — turn off the one the node doesn't use	sACN universe 2, or Art-Net 0-0-1
Faders	Fixtures	
Console 1–12	C42 #1–12 FOH	
Console 13–24	Zoom #13–18 LX1 · #19–20 LX2 · #21 SR · #22 SL · COB #23–24 LX2	
Wing 1 25–36	COB #25–28 LX2 · #29–30 SR · #31–32 SL · PixBar #33–36	
Wing 2 37–48	PixBar #37–38 · COB #39 · hazer #40 · pinspots #41–42 · 43–48 spare	

Both wings must be connected (fixtures 25–42). Printed fader labels:

03_Lighting_Mantra/TLM_R13_1_Mantra_Labels.pdf.

4 Step-by-step patch procedure

1 Back up what is on the desk

1. FAT32 USB stick in the Mantra. Home > Tools > Export Show — keep it untouched as the venue restore point.
2. Keep BASE_SHOW_2026.mtr (the clean venue base) on the stick too.

2 Load the R13.1 show file

1. Copy TLM_SHOW_2026_R13_FLASHY_SCENE_SPLIT.mtr to the stick (check it in Mantra Editor first — Part D §9).
2. Home > Tools > Import Show > choose it. Check five custom fixture types: CX 42 NEW, ZOOM 12 CHANNEL, TOURCOB PAR, PIXBAR 6CH, HAZER 2CH.

3 Clear the output

1. A L, then O, so nothing is live while you check the patch.

4 Check fixtures 1–40

1. Tools > Setup > Patch. Step through against section 1: number, library, universe, address.
2. Expected: #1–12 U2:101–232 · #13–22 U1:301–420 · #23–32 U1:421–480 · #33–38 U1:1–36 · #39 U1:481–486 · #40 U1:487–488.

5 Pinspots — only if the mirror ball is used

1. Generic dimmer, U1:37 as #41 and U1:38 as #42 → Patch.
2. Record both at full on **P5 M7** (free in R13.1). Save.

6 Network and universe 2

1. Tools > Setup > Network: 2.0.0.1, 255.0.0.0; turn off Art-Net or sACN (whichever the node doesn't use).
2. Set the node, connect it through the switch, run its DMX out to the FOH bar. Test C42 #1 at 50 %.

7 Remote trigger for QLab (R13.1)

1. Tools > Setup > Remote Triggers > Add > OSC · Play Memory · port 8000. Save.

8 Cable, power, terminate

1. One chain per universe in the order of section 3; label both ends; terminate the last fixture.
2. No more than about 1,840 W per 10 A circuit.
3. FOH bar: the C42s are powered from the venue dimmers — set those dimmer channels to NON-DIM (hard power) before switching on. Never dim an LED fixture.

9 Rig check and save

1. Each fixture alone at 50 %, then 0 → 100 %, then each colour. Use the ROW memories (100–103) for each type.
2. Confirm PixBar #38 and TourCOB #39 physically exist (R13.1 fix list). Save (T S), export to USB, tick the rig sheets.

5 Rig sheets by position

FOH SR end

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
41	PIN	1 ch	U1:37	Mirror ball (optional, not patched)	☐	☐	☐	☐

FOH SL end

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
42	PIN	1 ch	U1:38	Mirror ball (optional, not patched)	☐	☐	☐	☐

LX1

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
13	ZM	12 ch	U1:301	Top/front colour wash — DSR	☐	☐	☐	☐
14	ZM	12 ch	U1:313	Top/front colour wash — DSC	☐	☐	☐	☐
15	ZM	12 ch	U1:325	Top/front colour wash — DSL	☐	☐	☐	☐
16	ZM	12 ch	U1:337	Top/front colour wash — CSR	☐	☐	☐	☐
17	ZM	12 ch	U1:349	Top/front colour wash — CSC	☐	☐	☐	☐
18	ZM	12 ch	U1:361	Top/front colour wash — CSL	☐	☐	☐	☐

LX2

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
19	ZM	12 ch	U1:373	V3 Theodore — 13°, barn doors, steep top light	☐	☐	☐	☐
20	ZM	12 ch	U1:385	V4 Marina — 13°, barn doors, steep top light	☐	☐	☐	☐
23	COB	6 ch	U1:421	Backlight — DSR	☐	☐	☐	☐
24	COB	6 ch	U1:427	Backlight — DSC	☐	☐	☐	☐
25	COB	6 ch	U1:433	Backlight — DSL	☐	☐	☐	☐
26	COB	6 ch	U1:439	Backlight — CSR	☐	☐	☐	☐
27	COB	6 ch	U1:445	Backlight — CSC	☐	☐	☐	☐
28	COB	6 ch	U1:451	Backlight — CSL	☐	☐	☐	☐
33	PIX	6 ch	U1:1	Effect row, tilted downstage off the back wall (R-25)	☐	☐	☐	☐
34	PIX	6 ch	U1:7	Effect row, tilted downstage off the back wall (R-25)	☐	☐	☐	☐
35	PIX	6 ch	U1:13	Effect row, tilted downstage off the back wall (R-25)	☐	☐	☐	☐
36	PIX	6 ch	U1:19	Effect row, tilted downstage off the back wall (R-25)	☐	☐	☐	☐
37	PIX	6 ch	U1:25	Effect row, tilted downstage off the back wall (R-25)	☐	☐	☐	☐
38	PIX	6 ch	U1:31	Effect row, tilted downstage off the back wall (R-25)	☐	☐	☐	☐
39	COB	6 ch	U1:481	Extra backlight — centre (confirm fitted)	☐	☐	☐	☐

SR boom

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
21	ZM	12 ch	U1:397	High side — top of boom (~2.8 m), across to SL, head height	☐	☐	☐	☐
29	COB	6 ch	U1:457	Mid side — boom arm ~1.2 m, knee to hip	☐	☐	☐	☐
30	COB	6 ch	U1:463	Low side (shin) — boom arm ~0.4 m	☐	☐	☐	☐

SL boom

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
22	ZM	12 ch	U1:409	High side — top of boom (~2.8 m), across to SR, head height	☐	☐	☐	☐
31	COB	6 ch	U1:469	Mid side — boom arm ~1.2 m, knee to hip	☐	☐	☐	☐
32	COB	6 ch	U1:475	Low side (shin) — boom arm ~0.4 m	☐	☐	☐	☐

Floor, US

#	Type	Mode	Start	Role / focus	Addr	DMX	Pwr	Test
40	HAZE	2 ch	U1:487	Atmosphere — P5 M4 HAZE (50 %, fan 50 %)	☐	☐	☐	☐

6 Power, labels and troubleshooting

Position	Fixtures	Known load	Suggested
FOH	12 × C42	3,600 W (15.7 A)	3 FOH dimmer circuits, NON-DIM, 4 C42 each
LX1	6 × Zoom	~1,080 W (4.7 A)	1 × 10 A
LX2	2 Zoom, 7 COB, 6 PixBar	360 W + label ratings	1 × 10 A
SR / SL boom	1 Zoom + 2 COB each	180 W + labels	1 × 10 A each
Floor	Hazer	From label	1 × 10 A

Symptom	Likely cause	Fix
Two fixtures respond together	Duplicate address	Check displays against the master patch
Colours on the wrong channel	Wrong DMX mode	Set 11 / 12 / 6 ch to match section 2
All C42s dead, U1 fine	Universe 2 not reaching the node	Switch, node IP, node universe
Nothing past a fixture	Broken cable / through-port	Swap the cable after the last working unit
Random flicker	No terminator or mic cable	Terminate; use DMX cable
Fixture strobes or resets	Control/Strobe/Reset not 0	Clear it; record 0
Looks stack with odd colours	Another memory still playing (LTP)	O (All Cues Off) and re-fire; check the QLab release cue

Patch notes / changes